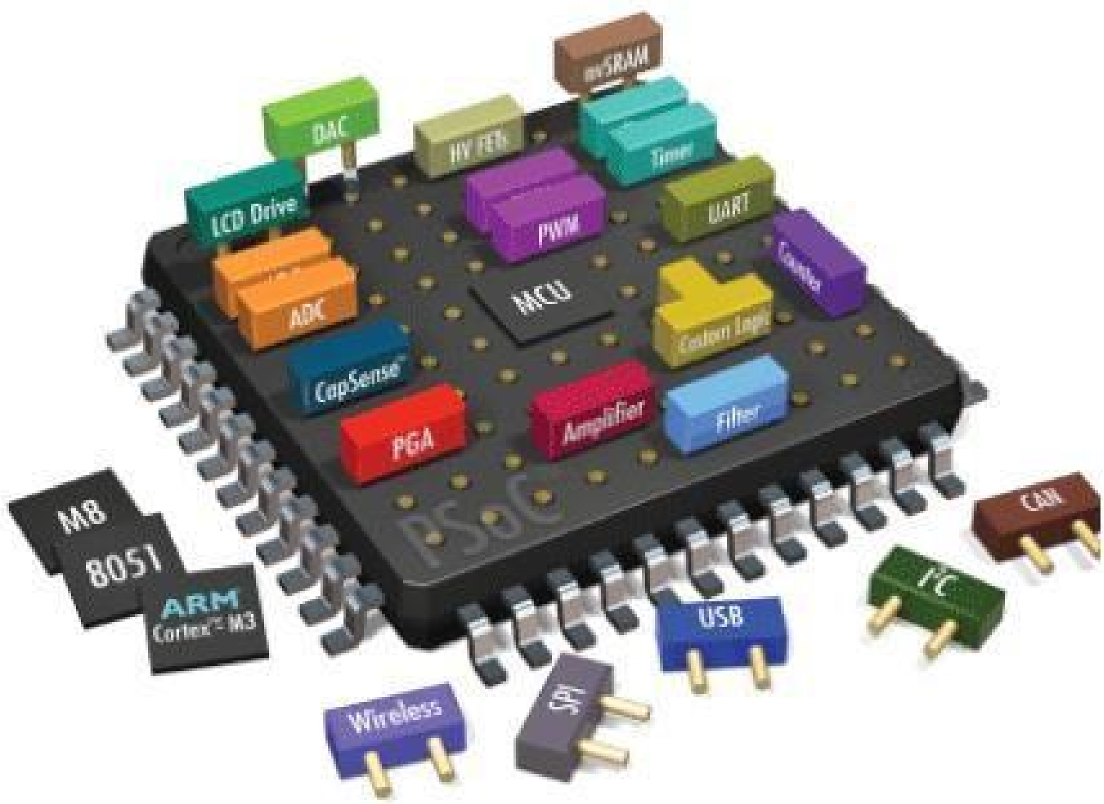
HITEC UNIVERSITY, TAXILA

DEPARTMENT OF COMPUTER ENGINEERING

Lab # 7: Demonstration and practice of Dot Matrix LED Control and display different patterns.

Embedded Systems

Demonstration and practice of Dot Matrix LED Control and display different patterns.



Dated:

Week 7

Semester:

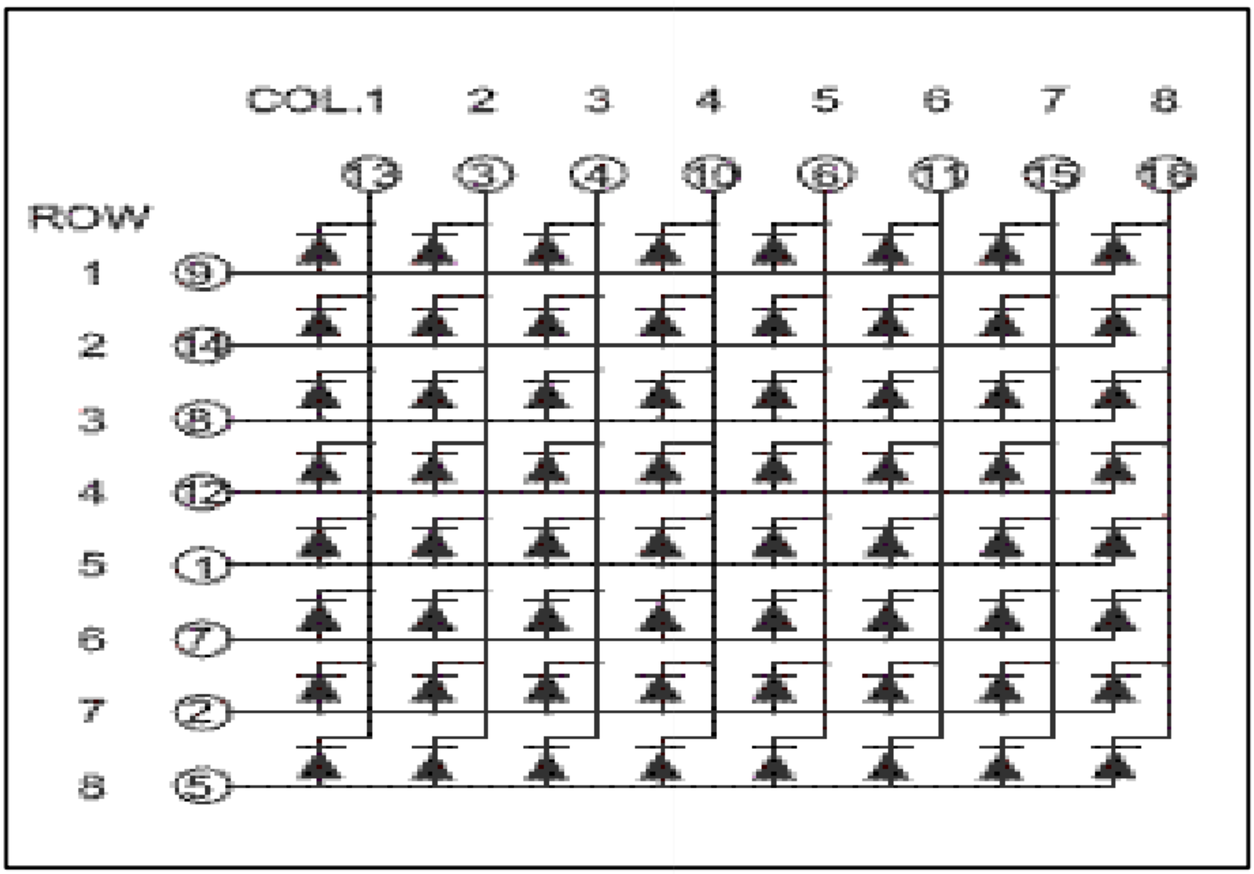
Fall 2023

Objective

1. To demonstrate that port 2 and 0 controls 8 8 dot matrix LED on the MTS-51 trainer.
2. To understand how to use 74LS245 and 2803 for controlling the rows and columns signals respectively

Circuit Description

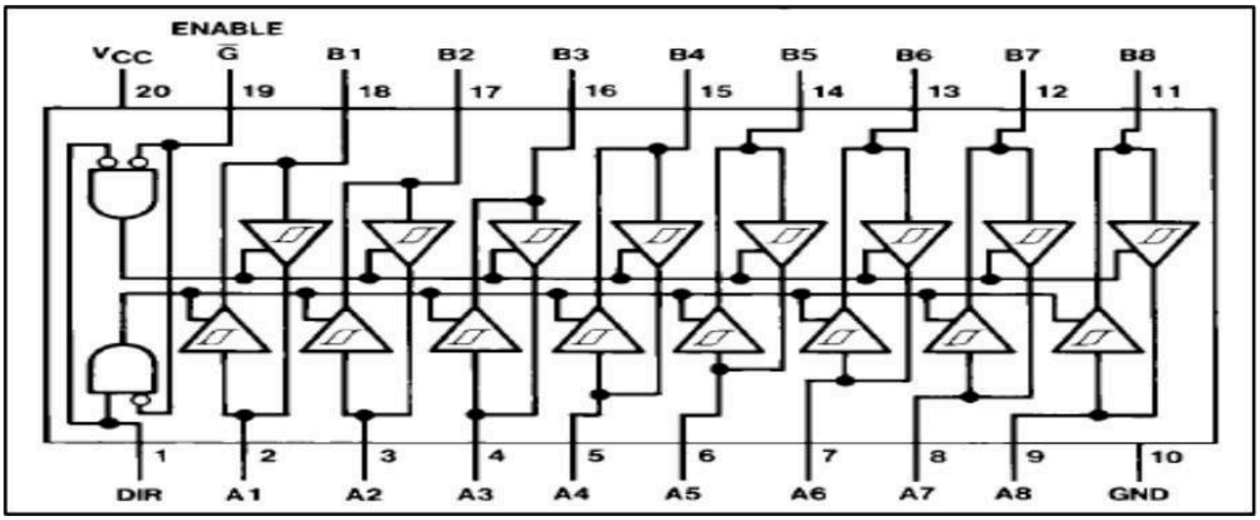
D3 is a common cathode matrix LED (all LED Cathodes are connected to the column line) as shown in Figure #1 below. When a row signal is high and a column signal is low, the corresponding LED is turned on.



The row scan signal is emitted from P2 lines through IC (74LS245) and the column scan signal is emitted from P0 through IC (2803).

Both row and column scan signal are active-high signals. That is, all LEDs are turned on when the row and column signals are all 1s. The 2803 Darlington pairs are used to sink column currents (Return to 2803 datasheet).

74LS245



The device allows data transmission from the A bus to the B bus or from the B bus to the A bus depending upon the logic level at the direction control (DIR) input (pin 1). The enable input (G) can be used to disable the device so that the buses are effectively isolated.

Function Table for 74LS245

Enable	DIR	Operation
L	L	B data to A bus
L	H	A data to B Bus
H	L or H	Isolation

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Exercise#1

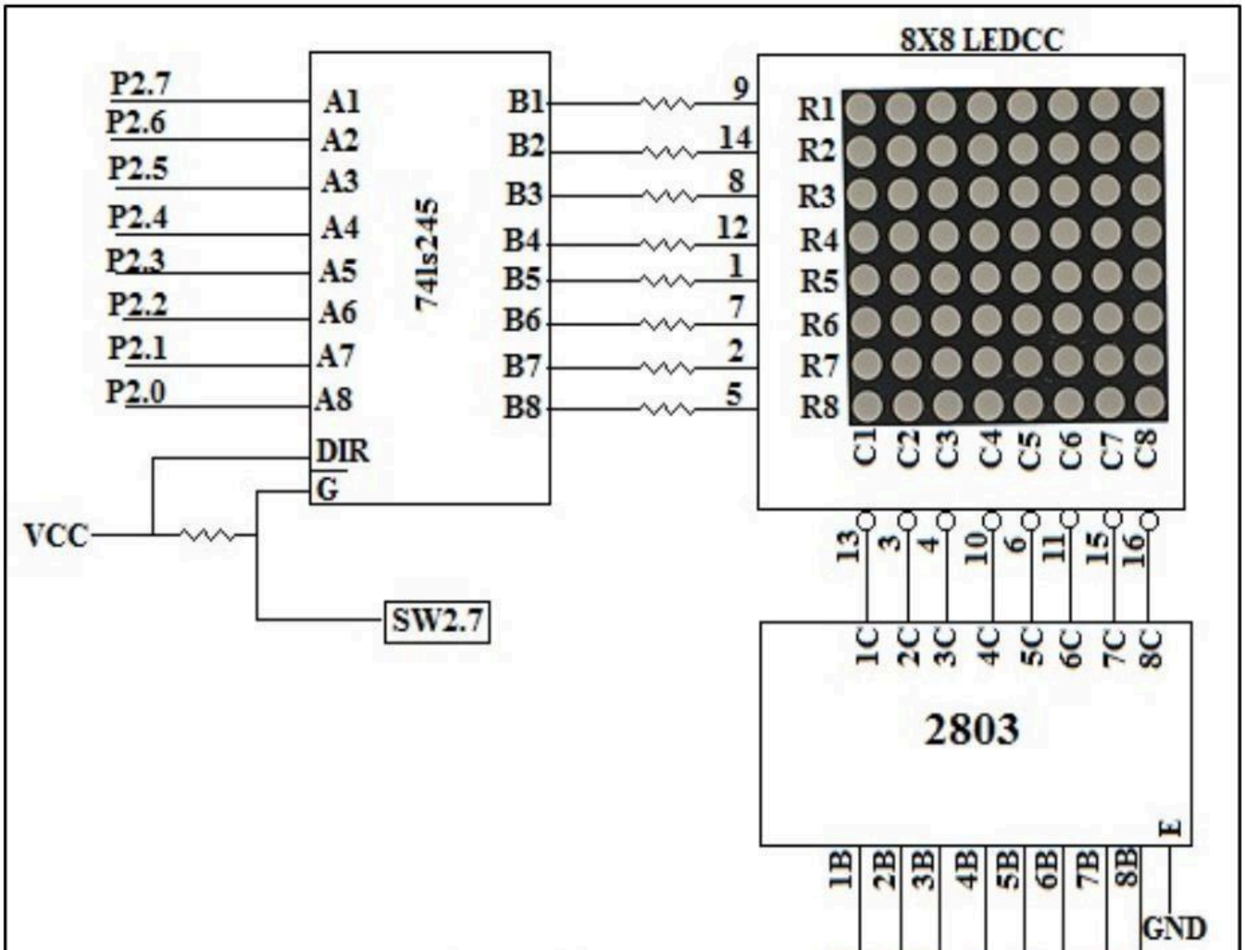

Figure #2

Functional Description:

Sequentially turn on the column LEDs from column 1 to column 8 eight time, and then turn on the row LEDs from row 1 to row 8 eight times.

Return to the introduction to follow the development of assembly program procedure.

Source Program

```
            ORG     000H

            CLR     P1.7
START:
            MOV     R0,#64
            MOV     P2,#FFH
            MOV     A,#10000000B
NEXT_COL:
            MOV     P0,A
            CALL    DELAY
            RR      A
            DJNZ    R0,NEXT_COL

            MOV     R0,#64
            MOV     P0,#FFH
            MOV     A,#10000000B
NEXT_ROW:
            MOV     P2,A
            CALL    DELAY
            RR      A
            DJNZ    R0,NEXT_ROW

            JMP     START

;=======================================
; DELAY 0.1S
;=======================================
DELAY:
            MOV     R6,#200
DL1:
            MOV     R7,#249
            DJNZ    R7,$
            DJNZ    R6,DL1
            RET
            END
```

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Program Description

Command	Description
CLR P1.7	When the matrix LED is controlled by program, this command must be used to float LCM data bus to prevent the P0 signal from electrical interference
MOV P2,#FFH	P2 emits FFH to row lines
MOV A,#10000000B	P0 emits a 1 to column 1(C1), then the LEDs in this column are turned on
RR A	The 1 is shifted to column 2 by rotate instruction to turn on the LED in C2.
MOV P0,#FFH	To turn on the LEDs row after row, P0 emits FFH to column lines
MOV P2,A	P2 emits a 1 to row 1 (R1), then the LEDs in this row are turned on
RR A	The 1 is shifted to row 2 by rotate instruction to turn on the LED in R2.

Lab Tasks:

1. Display the example in the manual and analyze the output.
2. Write a program which display the alphabet A on Dot Matrix LED.
3. Write a program which display the first alphabet of the name of all group members on Dot Matrix LED. Each alphabet stays on 0.5 seconds.

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Lab Performance Evaluation:

Read and practice the manual. Performance is evaluated at the end of lab based on successfully running and understanding of examples, tasks and viva using defined rubrics.

Lab Report Evaluation:

Submit the Lab Report by writing all the examples and tasks which must include the following:

1. Brief description about how its work (3-5 lines)
2. The code
3. The results (Screenshot)

Important note: Lab Report must be according to the Lab Report Format available on Google Classroom and must be submitted on time. Two similar reports will get zero marks. So, don't try to copy your fellow reports. Lab report must be submitted in group of three maximum.